

Hamza Saeed

Deep Learning & AI Engineer · Blockchain & Web3 Developer

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PROFILE

Deep Learning and AI Engineer specialized in building, training, and deploying advanced neural network architectures—including Vision Transformers (ViT), Swin Transformers, CNNs, and multi-agent LLM orchestration frameworks—to solve complex production problems. Brings extensive experience engineering core technical systems, decentralized applications, and robust cryptographic protocols. A rigorous engineer focused on driving technical innovation where intelligent automation, agentic workflows, and secure infrastructure converge to scale enterprise-grade ecosystems.

EDUCATION

Bachelor of Science in Computer Science · Lahore Garrison University Lahore | Dec 2022 – Jul 2026

- Relevant coursework: Deep Learning, Artificial Intelligence, Blockchain & Distributed Systems, Software Engineering, Database Systems, Computer Networks

EXPERIENCE

Community Ambassador · Deblock Remote | Sep 2022 – 2023

- Grew and managed an engaged blockchain community through AMAs, educational workshops, and consistent online presence.
- Forged strategic partnerships to expand Deblock's ecosystem reach and drive broader Web3 adoption.

Operational Head, Lead Generation · Zomgro Hasilpur | 2021 – 2022

- Directed structured outreach campaigns and data-driven strategies, delivering measurable growth in qualified lead volume.
- Applied data mining techniques to surface actionable insights and improve end-to-end operational efficiency.

AI & DEEP LEARNING PROJECTS

Multi-Agent AI Research Platform · Streamlit Cloud Remote | 2026

- Built a 4-agent autonomous pipeline (Search, Read, Write, Critique) using **LangChain** and Python to automate high-fidelity web scraping and information gathering.
- Designed a premium SaaS dashboard via Streamlit with custom CSS, featuring real-time telemetry execution logs, and automated multi-format compilation (**ReportLab** / **python-docx**).
- Reduced token overhead by 35% by developing a semantic data parser to prune boilerplate HTML DOM configurations from web sources before LLM context ingestion.
- Stack:** Python · LangChain · Streamlit · Tavily API · Gemini/Groq/OpenAI APIs · ReportLab · Git

Brain Tumor MRI Classification using Vision Transformer & Swin Transformer ·

- Developed a deep learning framework for automated brain tumor classification from MRI images using Vision Transformer (ViT) and Swin Transformer architectures.
- Applied transfer learning, image preprocessing, and data augmentation to improve model performance and generalization.
- Integrated Grad-CAM for Explainable AI to visualize model predictions and enhance interpretability.
- **Stack:** Python · PyTorch · Vision Transformer (ViT) · Swin Transformer · OpenCV · Scikit-learn · Grad-CAM

WEB3, BLOCKCHAIN & INFRASTRUCTURE PROJECTS

SeQureChain — Blockchain-Based Evidence Management DApp · LGU

Lahore | 2025 – 2026

- Deployed `EvidenceManager.sol` on Ethereum Sepolia to immutably anchor SHA-256 file hashes on-chain; integrated IPFS via Pinata for content-addressed storage to enable instant tamper detection.
- Replaced password authentication with MetaMask ECDSA wallet signatures for cryptographic non-repudiation; built a cross-platform frontend (React Native/Expo) backed by a Node.js/Express REST API.
- **Results:** 100% blockchain confirmation rate · 100% tamper detection · ~22s avg. upload · 50/50 concurrent uploads, zero failures.
- **Stack:** Solidity · Hardhat · Ethereum Sepolia · IPFS/Pinata · MetaMask · ethers.js · React Native · Expo · Node.js · Firebase · SHA-256

TECHNICAL SKILLS

Deep Learning & AI: Python, LangChain, PyTorch, TensorFlow, Keras, Vision Transformer (ViT), Swin Transformer, CNNs, Agentic Workflows, Transfer Learning, Computer Vision, Explainable AI (Grad-CAM), OpenCV, Scikit-learn, Generative AI APIs (OpenAI, Gemini, Groq), Tavily API

Blockchain & Web3: Solidity, Hardhat, Ethereum, IPFS, MetaMask (EIP-1193), ethers.js, Smart Contracts, DeFi Protocols, Tokenomics, Cryptographic Signatures, Contract Deployment & Unit Testing

Full-Stack Systems: Streamlit, React Native, Expo, Node.js, Express.js, Firebase, REST APIs, Git, GitHub

Research & Strategy: Deep Learning Research, Project Management, Web3 Architecture Strategy, DeFi Ecosystem Analysis, Crypto-economic Models, Protocol Evaluation, Technical Trend Analysis

CERTIFICATIONS

- Basics of Blockchain and web3
- Introduction to Decentralised Finance (DeFi)